Pronumerals, coefficients and algebraic expressions

1 For each of the following terms, choose the correct definition from the options below:

a Pronumeral **b** Coefficient **c** Expression

A A numeral that is placed before and multiplies a pronumeral in an algebraic term

B A way to express a mathematical value that involves pronumerals

C A symbol, commonly a letter, that is used in the place of a numeric value

2 State the coefficient in the following terms:

a $4x$ **b** $-4y$ **c** $6y$ **d** $-6y$

e $-2x^2$ **f** $8ab$ **g** $9z$ **h** $-11x^3$

i uv^2 **j** $\dfrac{4h}{3}$ **k** $-i$ **l** $\dfrac{-f}{5}$

Basic operations in algebra

3 For each of the following terms, choose the correct definition from the options below:

a Sum **b** Product **c** Difference **d** Quotient

e Factor **f** Term

A The result of multiplying two or more values or algebraic expressions.

B The result of dividing a value or algebraic expression by another.

C A single component of a product.

D The result of adding two or more values or algebraic expressions.

E A single component of a sum or subtraction.

F The result of subtracting a value or algebraic expression from another.

4 Write an algebraic expression for each of the following statements:

a 7 multiplied by x **b** 8 more than $5x$

c 3 less than $8y$ **d** The product of $9u$ and 8

g 8 more than the quotient of 9 and x

h The difference of a and b, divided by 6

5 Write a word statement for each of the following expressions:

a $8x$ b $p+7$ c $x-6$ d $\frac{p}{8}$

e $b-12$ f $15m$ g $\frac{u}{15}$ h $2(b+9)$

i $9y-3$ j $\frac{y}{3}+5$ k $12-3k$ l $\frac{m+n}{d}$

6 For each of the following expressions:

i State the order in which the operations should be performed.

ii Hence, simplify the expression.

a $3 \times u \div v + 8$ b $5 + y \times x - 4$ c $(4-k) \div 5$ d $7 - 12 \div (3 \times a)$

7 Rewrite in expanded form:

a $3k$ b $-5k$ c $2ab$ d $6x^2$

e pqr f $7m^2n$ g $-y^2$ h $(ij)^3$